

HORTA AP, PortugalWMO: **085050**Lat: **38.5211N**Lon: **28.7139W**Elev: **135**StdP: **14.62**Time Zone: **-1.00 (AZR)**Period: **94-13**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	50.0	51.6	39.0	35.2	52.8	40.7	37.6	53.8	34.0	57.8	31.1	58.1	11.1	0	0.408

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.2	78.8	71.9	77.3	71.0	76.6	70.5	73.7	76.3	72.9	75.6	72.1	74.8	9.9	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	122.8	75.3	71.9	118.4	74.6	71.3	116.0	74.2	37.2	76.4	36.5	75.7	35.8	74.8	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 28.6	25.4	22.6	46.4	81.6	1.9	1.8	45.1	82.9	44.0	84.0	42.9	85.0	41.5	86.3		
(5)			42.4	76.1	1.8	1.2	41.1	77.0	40.1	77.7	39.1	78.4	37.8	79.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	59.2	58.7	58.9	59.9	62.0	66.2	70.8	72.8	71.3	67.1	63.0	60.6		
	DBStd	5.70	3.07	2.89	2.79	2.49	2.46	2.35	2.42	2.26	2.49	3.13	3.35	2.90		
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0		
	HDD65	1036	181	175	188	153	96	14	0	0	0	15	77	137		
	CDD50	5203	285	245	277	297	373	487	644	706	639	531	390	329		
	CDD65	763	0	0	0	0	4	51	179	241	189	81	17	1		
	CDH74	1049	0	0	0	0	0	4	217	573	239	14	2	0		
	CDH80	10	0	0	0	0	0	0	1	8	1	0	0	0		
(14) Wind	WSAvg	10.6	13.0	12.8	12.7	11.0	10.1	8.8	7.2	7.9	9.2	10.6	11.6	13.0		
(15) Precipitation	PrecAvg	41.1	4.1	3.5	4.2	2.8	3.1	2.3	1.5	2.0	3.3	4.9	4.3	5.3		
	PrecMax	63.3	7.8	7.1	10.0	5.7	10.1	5.0	6.1	8.2	6.0	11.2	9.2	11.7		
	PrecMin	27.6	0.4	0.5	1.1	0.6	0.0	0.4	0.2	0.2	1.1	0.9	0.6	0.7		
	PrecStd	7.8	1.7	1.9	2.4	1.6	2.6	1.3	1.1	1.6	1.7	2.6	2.4	2.6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	66.0	65.9	66.4	68.1	69.9	74.3	79.1	80.8	79.1	75.5	73.3	66.6		
		MCWB	62.1	61.6	60.7	62.2	64.7	68.3	72.1	72.0	72.3	70.0	66.6	63.3		
	2%	DB	64.7	64.5	64.7	66.3	68.3	72.4	77.4	79.0	77.4	73.8	70.1	66.1		
		MCWB	61.6	60.7	60.0	61.6	63.6	67.1	70.7	71.7	71.1	69.2	66.5	63.4		
	5%	DB	64.2	63.2	63.9	64.8	67.5	71.4	76.3	78.2	76.6	73.0	68.8	65.1		
		MCWB	61.3	60.1	59.6	60.6	63.0	66.5	70.0	71.4	70.5	68.7	65.6	62.5		
	10%	DB	63.0	62.7	62.8	64.1	66.2	70.0	75.1	77.0	75.3	71.8	67.9	64.6		
		MCWB	60.6	59.8	59.2	60.1	62.3	65.5	69.3	71.0	69.5	67.8	64.8	62.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.8	62.9	62.9	64.6	66.6	70.8	74.6	75.2	74.2	72.5	69.5	65.9		
		MCDB	64.9	64.2	64.4	66.6	68.2	72.7	77.0	77.9	76.8	73.9	70.7	66.1		
	2%	WB	63.0	62.0	61.9	63.1	65.5	69.0	73.2	74.0	73.0	71.4	68.2	64.5		
		MCDB	64.0	63.3	63.5	65.0	67.2	71.0	75.6	76.7	75.6	72.8	69.3	65.2		
	5%	WB	62.3	61.4	61.1	62.1	64.6	68.1	72.1	73.3	72.2	70.3	67.0	63.7		
		MCDB	63.4	62.8	62.8	64.0	66.2	69.9	74.6	76.1	74.8	72.1	68.2	64.7		
	10%	WB	61.7	60.6	60.3	61.1	63.6	67.2	71.0	72.5	71.4	69.1	65.6	63.0		
		MCDB	63.0	62.2	62.2	63.1	65.4	69.2	73.7	75.5	74.0	71.2	67.3	64.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.9	6.4	6.6	6.9	7.1	7.2	8.4	8.2	7.8	7.3	6.6	5.8		
		MCDBR	5.6	6.3	6.9	7.4	7.8	8.1	9.2	8.9	8.4	7.4	7.1	5.3		
	5% WB	MCWBR	6.0	6.0	5.5	6.3	6.0	5.8	6.1	6.0	6.0	6.0	6.2	5.6		
		MCWBR	4.7	5.6	5.5	6.5	6.0	6.5	7.3	7.6	6.9	6.0	6.3	4.3		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.363	0.392	0.385	0.380	0.379	0.375	0.369	0.367	0.369	0.355	0.356		
		taud	2.502	2.483	2.383	2.429	2.473	2.509	2.533	2.571	2.576	2.552	2.536	2.478		
	Ebn at Noon		259	273	275	283	286	285	285	285	280	268	258	249		
		Edh at Noon	25	29	35	36	35	34	32	30	29	27	24	24		
(44) All-Sky Solar Radiation	RadAvg		638	906	1261	1642	1892	2013	2076	1958	1533	1068	717	549		
	RadStd		50	47	88	146	146	134	112	78	76	69	41	40		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Regional (3 neighbors)		N/A	N/A	+1.84	N/A	+0.63	+0.94	N/A	N/A	N/A	N/A

Nomenclature: See separate page